

Porting the Aurora Weather Model to Intel Accelerated Hardware

Iain Stenson, Rosie Wood, David Llewellyn-Jones, Tomas Lazauskas

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1 Title

Hi; I'm David Llewellyn-Jones, a Data Scientist at the Alan Turing Institute. I contributed to this work as part of the Research Computing Team alongside my colleagues Iain Stenson, Rosie Wood and Tomas Lazauskas, who is also here today.

I'd like to talk to you about some work we did together porting the Aurora Weather Model to Dawn.

2 Aurora: A Foundation Model for the Earth

Let me start by explaining what Aurora is. I'm not a meteorologist or climatologist so I can only give you a very high-level overview, but essentially Aurora is an AI weather model that's been trained on very large atmospheric datasets.

It's a **foundation model** which means it was trained on a very diverse set of data so that it's adaptable through fine-tuning to other more specialised forecasting tasks, such as forecasting air pollution.

Originally developed at Microsoft Cambridge, some of the team continued working on it at the Turing and that work is ongoing.

The model was developed using NVIDIA A100 80 GiB devices and we – as the Research Computing Team – were tasked with determining whether it could be moved successfully to non-NVIDIA platforms, in this case Dawn.

We wanted to find out how easy it would be to port, whether we'd get similar outputs, and how performance would compare.

3 Inference comparison

For inference accuracy we ran inference with a 7 day rollout in 6 hours steps, so here we see prediction results after 28 rollout steps for temperatures two meters above sea level across the entire world.

The top figure shows results for inference run on A100s. For this we used the Baskerville HPC based in Birmingham.

Below it are the results for the same inference steps executed on dawn.

Visually the results for Dawn and Baskerville look similar, which is exactly what we'd want.

But we were surprised to discover that despite what it looks like, even with the same configuration we are actually getting minor differences in the calculated results.

4 Inference error comparison

We can see this better if we plot the differences between these graphs.

This shows the error between the A100 and Dawn results.

What we can see is errors that range between 0 and 2 Kelvin and a Root Mean Square Error across the entire surface of 0.1 Kelvin.

This is despite the fact we have identical configurations and floating point precision in both cases.

5 Matrix-multiply-and-add errors

We were a little surprised by this so dug a bit into it.

There's a lot going on with the full model, so to simplify things to compare some basic matrix multiply and add operations.

We performed the operation 512 times for various widths of floating point operation on the GPU, comparing the results with the results from 64-bit floating point operations on the CPU.

As you might expect the errors accumulate over time.

What we found is that while BFLOAT 16 operations produce exactly the same results on both platforms, there are differences when using other precisions.

In particular the Intel GPUs gave consistently greater errors for FP 32 and consistently lower errors for FP 16 compared to the A100.

6 Intel GPU architecture

We were also interested in performance, but to understand our results I have to explain a little about the Intel GPU architecture.

Unlike for the A100, each Intel Data Center Max 1550 contains essentially two GPUs, connected together with a fast interconnect.

Each half has 64 GiB of memory, giving a total of 128 GiB when combined.

The driver can be configured to allow each of these halves to be addressed separately, called *flat* model, or they can be seen as a single graphics card, called *composite*.

In the latter case the processing units and memory are shared across the two halves.

7 Fine-tuning timings

With that explained, here we see some of our timing results for fine-tuning using the different systems and different hierarchy modes.

8 Fine-tuning timings for 1 half of a GPU

Let's first focus on these numbers.

These are the equivalent numbers for training with a single Torch device, whether that be a single half in the case of the flat hierarchy or both halves in the case of composite.

What we see is significantly improved results when using only half the GPU for the Intel case compared to using both halves in composite mode. Because the model fits on a single time, this is likely to be due to the impact of moving data across the interconnect.

Training on the Intel GPU is also half as fast as training on the A100. However this must be put into perspective.

9 Fine-tuning timings for a full GPU

Rather than leaving it to the driver, if we perform distributed data parallel, considering each half of the GPU as being a separate GPU and letting PyTorch handle memory management, we get much closer training speeds compared to the A100.

10 Fine-tuning time graph

In practice, what we see is that we have a very clear trade-off between model-size and performance.

With Aurora we can change the size of the model by increasing or decreasing the encoder and decoder depths.

What we see is that when using flat hierarchy on the Intel GPU we get roughly similar – only slightly worse – training times compared to the A100s.

However, we run out of memory for storing the model more quickly. The Intel GPU has 64 GiB of memory on each half. In composite mode we can combine this to give 128 GiB of memory in order to support much larger model sizes, but there's a really large performance hit from doing this.

11 Links

If you want to look further into the details we've written up a technical report which you can access with the link [here](#).

Aurora continues to be worked on and Iain is currently porting it to Jasmin, which offers predeployed weather datasets.

Weather models in general are large and getting larger, so the message I'm hearing is that providing more GPU memory as well as good documentation about CPU offloading – for example using DeepSpeed – would be a helpful direction for HPC to take for the Earth Systems Training community.